S01. Crystal-field and spin–orbit anisotropy in a finite ion model

Pure Thought research portfolio

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Source: synthesis challenge 11; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

An ion inside a crystal can respond differently to a magnetic field pointing in different directions. Its neighboring atoms shape the electron's available states, while the electron's spin interacts with its orbital motion. This problem asks whether a small, explicitly specified quantum model can establish those directional differences with reliable numerical bounds.

Key terms

- **Ion** — An atom or group of atoms carrying net electric charge.
- **Crystal field** — The influence of surrounding charges on an ion's electronic energy levels.
- **Spin** — An electron's intrinsic quantum angular momentum.
- **Orbital angular momentum** — Quantum angular momentum associated with spatial electronic motion.
- **Spin–orbit coupling** — An interaction linking spin and orbital angular momentum.
- **Hamiltonian** — The operator specifying the model's energies and quantum evolution.
- **Kramers doublet** — A pair of equal-energy states protected by time-reversal symmetry in this half-integer-spin setting at zero magnetic field.
- **g tensor** — A matrix describing the directional magnetic response of the selected effective doublet.
- **Excitation gap** — The energy separation from the chosen low-energy states to the next states.

Why this matters

Directional magnetic response is a basic ingredient in understanding magnetic materials and controlling spin states. A finite ion model lets us separate the roles of local geometry and spin–orbit coupling before tackling interactions among many ions.

What progress would mean

Progress would mean certified response ranges and excitation gaps over the specified parameter region, with clear limits on when the effective doublet description applies. This would establish a local mechanism; predicting a bulk magnet would require additional interactions and material information.

Revised core question

For a fixed d-shell occupancy and point-group representation, which anisotropy values are possible in a bounded crystal-field/spin–orbit Hamiltonian family?

Precise scope, target and outputs

Fix one ion model (start with d^1), a chosen point group such as C_4v, and a finite orbital-spin basis. Supply exact matrices for L,S and the group action. Define H(B)=H_CF+λL·S+μ_B B·(L+2S), with crystal-field coefficients and λ in a compact normalized box. Specify the temperature or isolated low-energy multiplet before defining susceptibility or the effective g tensor. For a Kramers doublet, extract a g tensor from the projected magnetic moment and use its singular values, which remove doublet-basis ambiguity. Require a certified excitation gap above that doublet. Optimize a bounded ratio or difference with a denominator bounded away from zero. Export matrices, symmetry identities, interval eigenvalue bounds, and global objective brackets.

Detailed critique of the synthesis framing

The synthesis's "all d^n configurations across every point group" has no interaction model, coefficient bounds, or objective normalization. Symmetry permits tensor components but does not determine their magnitudes; arbitrary crystal-field or spin–orbit scaling can make an optimization meaningless. "Rare-earth-like" is a motivation, not an observable. Many-electron d^n ions also require Coulomb/Hund parameters and a specified approximation. The revision retains an anisotropy atlas but begins with a finite, normalized spectral problem and separates a single-ion response from bulk magnetocrystalline anisotropy.

Validation and milestones

Baseline: exact symmetry-allowed Hamiltonian basis and one doublet g tensor. Strong: certified parameter-box extrema with a doublet-gap condition. Extend to d^2 only after specifying interactions. Check the isotropic limit, group covariance, and basis invariance of g singular values.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$H(B) = H_{\mathrm{CF}} + \lambda L \cdot S + \mu_B B \cdot (L + 2S),$$

$$PH(B)P = E_0 I + \frac{\mu_B}{2} \sum_{a,b} B_a g_{ab} \sigma_b + O(|B|^2).$$

Use the isolated Kramers-doublet projector P and exact orbital-spin conventions. The singular values of g remove doublet-basis ambiguity. Certify the gap above the doublet and bound any denominator in an anisotropy ratio on the compact parameter box. This finite ion model does not establish bulk magnetism.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.